EVID: Reference-free Evaluation of Action Fidelity in Videos from Robot World Models via Inverse Dynamics

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Abstract

Action-conditioned video world models predict the video a robot would record when it executes given actions, and they are evaluated mostly by how closely their frames match the recorded video. Such appearance metrics need the real video, which does not exist for new actions, and do not check whether the robot performs the commanded motion. Reproducing IRASim and its baselines and adding Cosmos-Predict2.5 on 105 Bridge and 57 Language-Table clips, we find that SSIM rates a still video as high as a correct generation and above every generation from corrupted actions. We propose EVID, which closes the loop from actions to video and back: an off-the-shelf inverse dynamics model reads the end-effector trajectory from the generated video, EVID measures its distance to the commanded trajectory, and inverse kinematics checks that the real arm could execute it. Without any real video, EVID separates real videos from a still one by a factor of six, grows monotonically with the corruption, and ranks the models as the reference-based metrics do. It shows that the action input, not the quality of the frames, decides whether the robot follows the commands, while the most frequent remaining errors are objects that change or vanish. Code, videos, and a results viewer are available at hoangducanh1865.github.io/PI.

1. Introduction

A video world model for robot manipulation predicts the video $\hat{V}$ a robot would record from its first frame I_0 and commanded actions $a_{1:T}$; if accurate, it can replace a physics simulator for policy evaluation, planning, and data generation [17, 27, 31]. Such models are compared mostly with frame-level appearance metrics against the recorded video, PSNR, SSIM [26], and LPIPS [30], or with FVD [24] over sets of videos. These metrics need the real video, which a simulator does not have for new actions, and they score how frames look, not whether the robot performs the commanded motion or whether a real arm could.

Regenerating 105 Bridge [25] clips with IRASim [31] from deliberately corrupted actions confirms the gap: SSIM rates a still video as high as the correct generation and above every corrupted one, whereas an end-effector track compared with the real video degrades steadily. What a simulator always has is the commanded actions, so we check the video against them. Inverse dynamics models have been used to read actions from generated videos and execute them, scoring task success [4, 23, 29]; such scores need a matched simulator or robot and mix the errors of the reader with those of the world model [14]. We instead compare the trajectory read from the video directly with the commanded one and measure the reader's own error on real videos.

Our main contributions are: (1) an evaluation framework that lists what each metric group measures and cannot measure, with a perturbation test of whether a metric notices wrong actions, and (2) EVID, a reference-free metric of action fidelity with an executability check, validated on real and corrupted videos and applied to IRASim, its baselines, and Cosmos-Predict2.5.

2. Related work

World models and inverse dynamics. UniSim [27] learns a simulator of real-world interaction from mixed data and trains policies inside it. IRASim [31] generates robot videos from a first frame and end-effector actions, conditioning each frame on its own action, and is used for policy evaluation with human-judged rollouts and for planning. Cosmos [17] provides video world foundation models, one of them post-trained on Bridge with actions. Inverse dynamics models, which predict the action between two frames, label unlabelled videos in VPT [1], turn video plans into robot actions in UniPi [3], and label generated videos as training data in DreamGen [9]; these works trust the actions read, while we compare them with the commanded ones.

Evaluating generated videos. VideoPhy [2] and WorldModelBench [11] judge physical commonsense and instruction following; Physics-IQ [16] compares predicted continuations of real experiments with the recorded ones and finds that realism does not imply physical understanding; EWMBench [28] compares end-effector trajectories with the ground truth and notes that static videos score high on scene consistency. WorldSimBench [23], World-in-World [29], and WoW-World-Eval [4] read actions from generated videos and execute them in simulators or on robots, scoring task suc-

cess; KineBench [14] replaces the inverse dynamics model by pose estimation because its errors confound the result; SIMPLER [12] matches simulators to real setups, including Bridge. EVID needs neither the real video nor a simulator, and it measures the error of its reader to separate it from that of the world model. Table 4 in the appendix compares the main sources.

Open questions. Does a realistic frame imply that the robot followed the commanded actions, do objects move only when touched and keep their identity, and could the real arm execute the motion? We measure the first and the third with EVID, and the second only through vision-language judges.

3. Reproducing IRASim

Pipeline. IRASim [31] predicts 16 frames from the first frame and 15 end-effector actions, each the motion between two frames in the end-effector frame of the earlier one, plus the gripper state. The SDXL [22] VAE encodes each 256×320 frame into a $4\times32\times40$ latent, and a diffusion transformer [21] denoises the latents of frames 1 to 15 with the clean latent of frame 0 as the condition; it is trained with the noise-prediction loss [6] and sampled with 50 PNDM [13] steps. Actions enter through adaptive layer norm: Frame-Ada modulates each frame by its own action and Video-Ada every frame by one embedding of the sequence, while the baselines LVDM [5] and VDM [7] condition on the whole sequence in latent and in pixel space.

Setup. We use the authors' code, checkpoints, and evaluation settings on 105 Bridge [25] test clips of 16 frames at 5 Hz, the first episodes of each of 12 instruction verbs, and on 57 Language-Table [15] clips. Cosmos-Predict2.5 without actions, at 3 min per clip on an A100, runs on a fixed subset of 24 of the Bridge clips, and all Cosmos videos are resampled to 16 frames at 256×320. The initial noise of a clip depends on its batch, so seeds are samples, and Frame-Ada is the mean of two.

Differences from the paper. We do not compute FVD [24], which gives no per-clip score and ignores the commanded actions. Our PSNR is lower than reported, with the same ranking of models (Tab. 2). The authors' code scores the same frames against the same real videos, and on 64 test clips drawn at random Frame-Ada reaches 24.85 dB, within 0.43 dB of the reported 25.28, so the first episodes of each verb that we score are harder than the average test clip.

4. Evaluation framework

Two references bound every score: the real video of the same actions is the ceiling, and the still video, the first frame held for 16 frames, is the floor, with perfect image quality and no motion. Cosmos-Predict2.5 [17] enters with actions, post-trained on Bridge, and without actions, given only the first frame and the instruction, which tests whether the action input matters. Table 1 lists the metric groups; all are computed per clip, and we call a difference between two models significant when a paired Wilcoxon signed-rank test over the clips gives $p < 0.05$. The 2-D track follows the moving CoTracker3 point best explained by the recorded end-effector position, confirmed to lie on the gripper by two vision-language models, and compares its generated and real tracks with $\mathrm{NDTW} = \exp(-\mathrm{DTW}/(16\,d_{\mathrm{th}}))$, $d_{\mathrm{th}} = 20$ px.

5. EVID: reading the actions back

Idea. A simulator always has the commanded actions, so EVID checks the video against them (Fig. 1): an inverse dynamics model reads the end-effector trajectory back from the generated video, and we measure how far it deviates from the commanded trajectory and whether the arm can execute it. A video that shows the commanded motion closes this cycle with a small error. We use NVIDIA Cosmos3-Nano [18] in its inverse dynamics mode for the Bridge robot, without training.

From the model output to a trajectory. Let T_t be the 4×4 pose of the end-effector at frame t. For each pair of frames the model returns 10 numbers $\tilde{a}_t$ encoding the motion $\Delta_t = T_{t-1}^{-1} T_t$, three for the translation, six for the rotation, and one for the gripper, each scaled by NVIDIA's encoder [19] with the published 1st and 99th percentiles q_{01}, q_{99}:

$$\tilde{a} = 2\,\frac{a - q_{01}}{q_{99} - q_{01}} - 1, \qquad \hat{T}_t = \hat{T}_{t-1}\,\hat{\Delta}_t, \quad \hat{T}_0 = T_0. \tag{1}$$

We invert the scaling and chain the motions from the true first pose; the positions of $\hat{T}_{1:15}$ form the trajectory $\hat{p}_{1:15}$. Nothing is fitted: recorded states passed through NVIDIA's encoder and our decoder return within $2 \cdot 10^{-4}$ mm, once a roll offset of π between IRASim's and NVIDIA's Bridge states is applied.

EVID error and executability. The EVID error of a clip is the mean distance between the positions read from the video and the commanded ones,

$$E = \frac{1}{15} \sum_{t=1}^{15} \|\hat{p}_t - p_t\|_2 \text{ (cm)}. \tag{2}$$

A clip is executable when the inverse kinematics of the WidowX 250 6-DoF arm, with the model, solver, and limits of its Interbotix driver, reaches every pose read from the video within the joint limits and below the joint speed limit of π rad/s between frames 0.2 s apart. Because the driver's solver rejects 4 of the 105 trajectories recorded on the real arm, we add more starting guesses and a search over solution branches; the check then accepts 104, and without the roll offset it rejects 1,669 of the 1,680 recorded poses. Since EVID needs no real video, it can also select the best of several samples generated from the same actions.

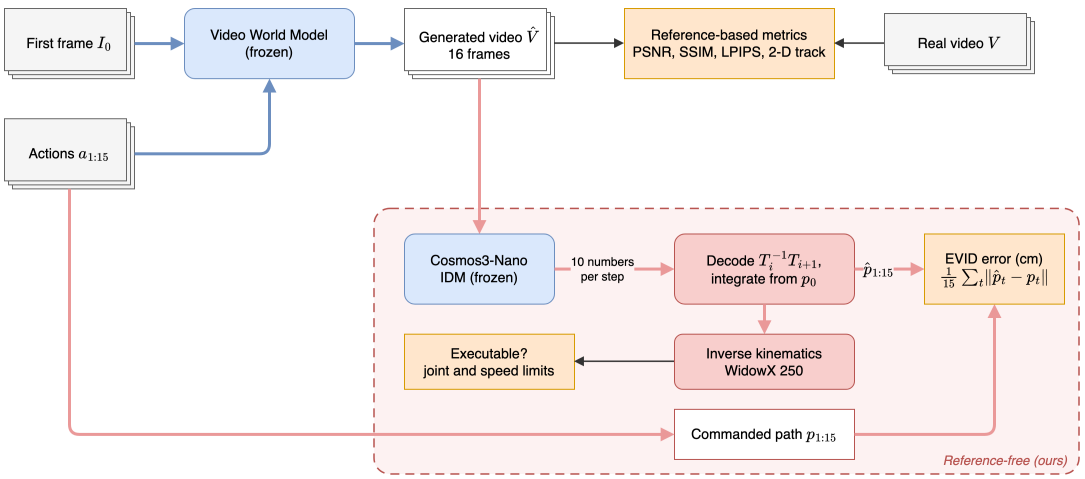


Figure 1. Evaluation pipeline. Both networks are frozen; reference-based metrics need the real video, while EVID (dashed box) needs only the commanded actions.

Group and metrics	Measures	Cannot measure
Appearance: PSNR, SSIM [26], LPIPS [30]	how close each frame looks to the recorded one	whether the robot moved as commanded: a still video keeps a high SSIM
Temporal: DINOv2 [20] frame similarity, warp error	smoothness and identity across frames	correct motion: the still video scores perfectly
2-D track: CoTracker3 [10] end-effector track, NDTW [8]	whether the gripper follows the image path of the real clip	depth and rotation; needs the real clip
Judges: two vision-language models	visible violations and whether the final state matches the real clip	a stable verdict: the judges differ in strictness
Action fidelity: EVID, ours	whether the 3-D trajectory read from the video matches the commanded one, and whether the arm could execute it	objects and contact; resolution bounded by the reader's error, 1.50 cm

Table 1. Metric groups. Task completion is covered by the outcome question of the judges.

	Appearance			2-D	EVID	Paper
Model	PSNR↑	SSIM↑	LPIPS↓	NDTW↑	cm↓	PSNR↑
Bridge, 105 clips						
IRASim-Frame-Ada	23.87	0.808	0.083	0.709	1.82	**25.28**
IRASim-Video-Ada	23.52	0.804	0.087	0.682	1.95	24.73
LVDM	22.62	0.788	0.103	0.593	2.54	23.55
Cosmos-Predict2.5	**25.74**	**0.859**	**0.063**	**0.754**	**1.64**	-
Real video	-	-	-	1	1.50	-
Still video	19.80	0.805	0.141	0.156	9.06	-
Language-Table, 57 clips						
IRASim-Frame-Ada	**27.15**	0.861	**0.059**	**0.702**	-	**28.82**
IRASim-Video-Ada	23.04	0.833	0.104	0.304	-	23.89
LVDM	26.44	**0.863**	0.066	0.646	-	28.25
Still video	22.35	0.879	0.103	0.131	-	-

Table 2. Main results, frames 1 to 15. 2-D: NDTW of the end-effector track, 96 Bridge clips; EVID: EVID error, no reader for Language-Table; Cosmos-Predict2.5 with actions; Frame-Ada: mean of two seeds (std at most 0.17 dB PSNR, 0.03 cm); Paper: PSNR reported in [31] on its full test set. Best in bold, second underlined, references in italics excluded.

	Appearance			2-D	EVID	
Actions	PSNR↑	SSIM↑	LPIPS↓	NDTW↑	cm↓	Exec.
Original	**23.95**	**0.809**	**0.082**	**0.699**	**1.81**	99.0
Noise 0.5 cm	21.65	0.782	0.107	0.557	2.57	99.0
Noise 1.5 cm	18.93	0.744	0.152	0.256	5.46	99.0
All zero	19.01	0.748	0.160	0.170	8.88	100
Mirrored	18.22	0.730	0.179	0.093	14.56	93.3
Still video	19.80	0.805	0.141	0.156	9.06	100

Table 3. Perturbation test: IRASim-Frame-Ada on the 105 Bridge clips from corrupted actions, scored against the real clip of the original actions. Exec.: % of clips executable by the WidowX 250, a necessary condition, not ranked.

6. Results

6.1 Reproduction and model comparison

Among the IRASim models, Frame-Ada leads on every metric, Video-Ada is second, and LVDM third, with significant differences on PSNR, 2-D NDTW, and EVID (Tab. 2); Language-Table gives the same order, except that Video-Ada falls far behind, as in the paper. Cosmos-Predict2.5 with actions has the best mean on every metric and a significantly higher PSNR, 1.9 dB above Frame-Ada; its EVID error of 1.64 cm is within 0.14 cm of the error on the real videos, the floor set by the reader. SSIM hardly separates the IRASim models from the still video.

6.2 Do the metrics notice wrong actions?

We regenerate the 105 clips with Frame-Ada from corrupted actions (Tab. 3). NDTW and EVID degrade steadily with the



(a) "put eggplant in pot": success
PSNR 23.85, EVID 1.04 vs 1.00 cm

(b) "put the toy in the vessel": the toy vanishes
PSNR 24.01, EVID 3.28 vs 2.52 cm

(c) "wipe pot with sponge": the sponge breaks up
PSNR 21.90, EVID 2.59 vs 1.80 cm

Figure 2. Frames 0, 8, and 15 of the real video (top) and of IRASim-Frame-Ada from the same actions (bottom); both judges agree on each verdict. EVID errors of the generated and the real video. Both object failures raise the EVID error by less than 1 cm, and PSNR rates (b) highest: the judges see these failures, neither metric does.

corruption, place zero actions at the level of the still video, and mirrored motion beyond it; EVID does so without the real video. SSIM rates the still video above every corrupted generation and as high as the correct one, and PSNR and LPIPS prefer it to the three strongest corruptions.

6.3 Validating EVID

On real videos the EVID error is 1.50 cm (95% interval 1.36 to 1.68) for clips that move 25 cm on average, while a still video scores 9.06 cm, as poorly as assuming no motion, so the reader uses the frames rather than a prior. EVID grows monotonically with the corruption and ranks the models as the reference-based metrics do, each step being significant. The executability check accepts 99% of clips for every action-conditioned model, the only failure being a clip the reader misreads on the real video as well, and rejects 7% of the mirrored ones: the arm motion of these models is almost always feasible, so executability is a necessary condition, and the EVID error is what separates the models.

6.4 What EVID reveals

Without actions, Cosmos-Predict2.5 produces smooth videos, with a DINOv2 adjacent-frame similarity of 0.916 against 0.952 for real ones, whose robot does not follow the recorded motion: on the 24-clip subset its EVID error of 10.30 cm does not differ significantly from the still video's 9.52 cm, against 1.74 cm with actions, and it can even replace the WidowX by another robot arm. Two vision-language judges, gemini-3.8-flash and gpt-6.1-sol, differ in strictness, flagging 32% and 1% of real clips; for every IRASim model they most often flag objects that change or vanish, in 47% and 10% of Frame-Ada clips, which neither EVID nor the appearance metrics detect (Fig. 2). Keeping the sample with the lowest EVID error among six Frame-Ada seeds raises PSNR from 23.92 to 24.07 dB and the 2-D NDTW from 0.720 to 0.736, small but significant gains on metrics that use the real video, which the selection does not; for PSNR this is a sixth of what an oracle selecting with the real video reaches.

7. Strengths and limitations of IRASim

Strengths. *S1, the robot follows the actions:* the EVID error of Frame-Ada is only 0.32 cm above that of the real videos. *S2, per-frame conditioning helps:* with the same backbone, Frame-Ada beats Video-Ada on every metric. *S3, stable:* its seeds do not differ significantly, and a rerun with the same batches reproduces the videos exactly.

Limitations. *L1, objects lose their identity,* typically once grasped (Fig. 2b). *L2, correct motion does not imply the correct outcome:* only 51% and 77% of Frame-Ada clips end in the real state according to the two judges. *L3, errors grow with contact:* the EVID gap to the real video is largest for wipe, turn, slide, and topple, 0.5 to 0.7 cm, and close to zero for open, flip, and fold. *L4, short clips of one embodiment:* 3 s at 256×320 per checkpoint.

8. Conclusion

Frame-level appearance metrics cannot tell whether a generated robot video follows its actions; EVID can, without a real video, by reading the actions back with an off-the-shelf inverse dynamics model decoded with its authors' formulas. On Bridge it shows that the action input matters more than the quality of the frames, that the arm motion of current models is almost always executable, and that objects are their most frequent visible failure. Its limits are one embodiment, 105 clips harder than the average test clip, judges that disagree, and a reader whose own error of 1.50 cm bounds what EVID resolves, which may have seen Bridge test episodes and may misread blurred generated frames; replaying the read motion in a simulator such as SIMPLER [12] would add dynamics and contact.

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A. Literature matrix

Source	Inputs	Outputs	Evaluation	Assumptions	Limitations
Video world models					
UniSim [27]	observation, language instruction or low-level control	video of the visual outcome	FID, FVD, IS, CLIP score in ablations; policies trained purely in the simulator and deployed zero-shot on real robots	mixing internet, simulation, and robot data yields one simulator of real interaction	hallucinated outcomes for unrealistic actions and limited memory, as stated by the authors
IRASim [31]	first frame, end-effector action trajectory	16 frames at 256×320	PSNR, SSIM, latent L2, FID, FVD against recorded videos; human preference; policy evaluation on LIBERO with human-judged success, Pearson 0.99 with the simulator; model-based planning	closeness to the recorded video measures simulation quality	action following is checked by frame metrics or by humans judging task success; one embodiment per checkpoint
Cosmos [17]	video and text; actions after post-training	video	3-D consistency by epipolar error and pose estimation; physics alignment by PSNR, SSIM, DreamSim, IoU on simulated scenes	pretraining on large video data transfers to physical AI after post-training	physics alignment measured on simulated scenes
Video models turned into actions by inverse dynamics					
UniPi [3]	first frame, text goal	video plan, then actions from an inverse dynamics model	task success in simulated manipulation; on Bridge, CLIP score, FID, FVD, and a success classifier on the last generated frame	a generated plan is executable and its actions can be recovered	slow diffusion sampling, as stated by the authors; plan errors reach the robot unchecked
DreamGen [9]	first frame, language, target robot	video, then pseudo-actions from a latent action or inverse dynamics model	DreamGen Bench with instruction following and physics alignment by VLM and human judges; downstream policy success	pseudo-actions from generated videos are useful training data	the benchmark scores the video, not the recovered actions
Evaluation of generated videos and of policies					
EWMBench [28]	first frame, language instruction	robot manipulation video	scene consistency, end-effector trajectory against the ground truth, semantic alignment	the recorded video and its trajectory are the reference	needs the real video; a static video can score high on scene consistency, as the authors note
WorldSimBench [23]	first frame, instruction	video, then actions from a video-to-action model	human-preference evaluator; task success of the actions executed in simulators (Minecraft, CARLA, CALVIN)	a generated video is good if the actions it implies succeed	needs a simulator of the task; errors of the video-to-action model mix with those of the video
KineBench [14]	generated video	6-D end-effector poses from vision foundation models, replayed in ManiSkill3	closed-loop task success and kinematic metrics on 20 tasks	explicit pose estimation avoids the errors of a learned inverse dynamics model	simulated tasks only; orientation estimates limit precision, as the authors note
VideoPhy [2]	text prompt	text-to-video generation	human judgement of semantic adherence and physical commonsense; automatic judge VideoCon-Physics	physical commonsense can be judged from the video alone	no actions or robots; best model correct on 39.6% of prompts
Physics-IQ [16]	frames of a real physical experiment	predicted continuation	spatial and spatiotemporal IoU, weighted spatial IoU, and MSE of motion against the recorded continuation	the real continuation is the reference	needs the real video; no robot actions
SIMPLER [12]	robot policy	rollouts in simulated scenes matched to real setups	correlation of simulated and real success rates	reducing control and visual gaps suffices without full digital twins	evaluates policies rather than videos; few real setups

Table 4. Literature matrix of the main sources. Evaluation methods are taken from each paper.